

2023-2024

Алгоритм ба Өгөгдлийн БҮТЭЦ - ЛАБ

io × UFE



```
class ArrayList {
  constructor() {
    this.items = new Array(100);
    this.size = 0;
  }

  get(index) {
    if (index < 0 || index >= this.size) {
      throw new Error('bad index');
    }
    return this.items[index];
  }

  insert(index, item) {
    // TODO
  }

  delete(index) {
    // TODO
  }

  getSize() {
    return this.size;
  }
}
```

```
class ArrayList {
  ...
  insert(index, item) {
    if (index < 0 || index >= this.size) {
      throw new Error('bad index');
    }
    // TODO
    this.size++;
  }
  ...
}
```

```
class ArrayList {
  ...
  delete(index) {
    if (index < 0 || index >= this.size) {
      throw new Error('bad index');
    }
    // TODO
    this.size--;
  }
  ...
}
```

**Insert, Delete method-
уудыг тодорхойлцгооё**

```
class Node {  
  constructor(value) {  
    this.value = value;  
    this.next = null;  
  }  
}
```

```
class LinkedList {  
  constructor() {  
    this.head = null;  
    this.size = 0;  
  }  
  
  get(index) {  
    if (index < 0 || index >= this.size) {  
      throw new Error('bad index');  
    }  
    //TODO  
  }  
  
  insert(index, item) {  
    // TODO  
  }  
  
  delete(index) {  
    // TODO  
  }  
  
  getSize() {  
    return this.size;  
  }  
}
```

```
class LinkedList {  
  ...  
  insert(index, item) {  
    if(index < 0 || index >= this.size) {  
      throw new Error('bad index');  
    }  
  
    if(this.head === null) {  
      this.head = item;  
      return;  
    }  
  
    let current = this.head  
    let i = 1;  
    while(i < index) {  
      current = current.next;  
      i++;  
    }  
  
    item.next = current.next;  
    current.next = item;  
  }  
  ...  
}
```

```
class Node {
  constructor(value) {
    this.value = value;
    this.prev = null;
    this.next = null;
  }
}
```

```
class DoubleLinkedList {
  constructor() {
    this.head = null;
    this.size = 0;
  }

  get(index) {
    if (index < 0 || index >= this.size) {
      throw new Error('bad index');
    }
    //TODO
  }

  insert(index, item) {
    // TODO
  }

  delete(index) {
    // TODO
  }

  getSize() {
    return this.size;
  }
}
```

```
class DoubleLinkedList {
  ...
  insert(index, item) {
    if(index < 0 || index >= this.size) {
      throw new Error('bad index');
    }

    if(this.head === null) {
      this.head = item;

      return;
    }

    let current = this.head
    let i = 1;
    while(i <= index) {
      current = current.next;
      i++;
    }
    item.prev = current.prev
    item.next = current.next;
    if(current.prev !== null) {
      current.prev.next = item;
    }
    if(current.next !== null) {
      current.next.prev = item;
    }
  }
  ...
}
```

```
class Queue {
  constructor() {
    this.size = 0;
    this.list = new LinkedList();
  }
  get(index) {
    if (index < 0 || index >= this.size) {
      throw new Error('bad index');
    }
    return this.list.get(index);
  }
  enqueue(item) {
    // TODO
  }
  dequeue() {
    // TODO
  }
  getSize() {
    return this.size;
  }
  getHead() { /* TODO */ }
  getTail() { /* TODO */ }
}
```

```
class Queue {
  ...
  enqueue(item) {
    this.list.insert(this.size, item);
  }
  dequeue() {
    return this.list.delete(0);
  }
  ...
}
```

```
class Stack {
  constructor() {
    this.list = new LinkedList();
    this.top = -1;
  }
  getTop() { /* TODO */ }
  push(item) {
    // TODO
  }
  pop() {
    if(this.top < 0) {
      return new Error('Stack is empty');
    }
    // TODO
  }
  getSize() {
    return this.size;
  }
}
```

```
class Stack {
  ...
  getTop() {
    return this.list.get(this.top);
  }
  push(item) {
    this.top++;
    this.list.insert(this.top, item);
  }
  ...
}
```

**Жагсаалт, дараалал, стекийн
үлдсэн method-уудыг гүйцээ.**